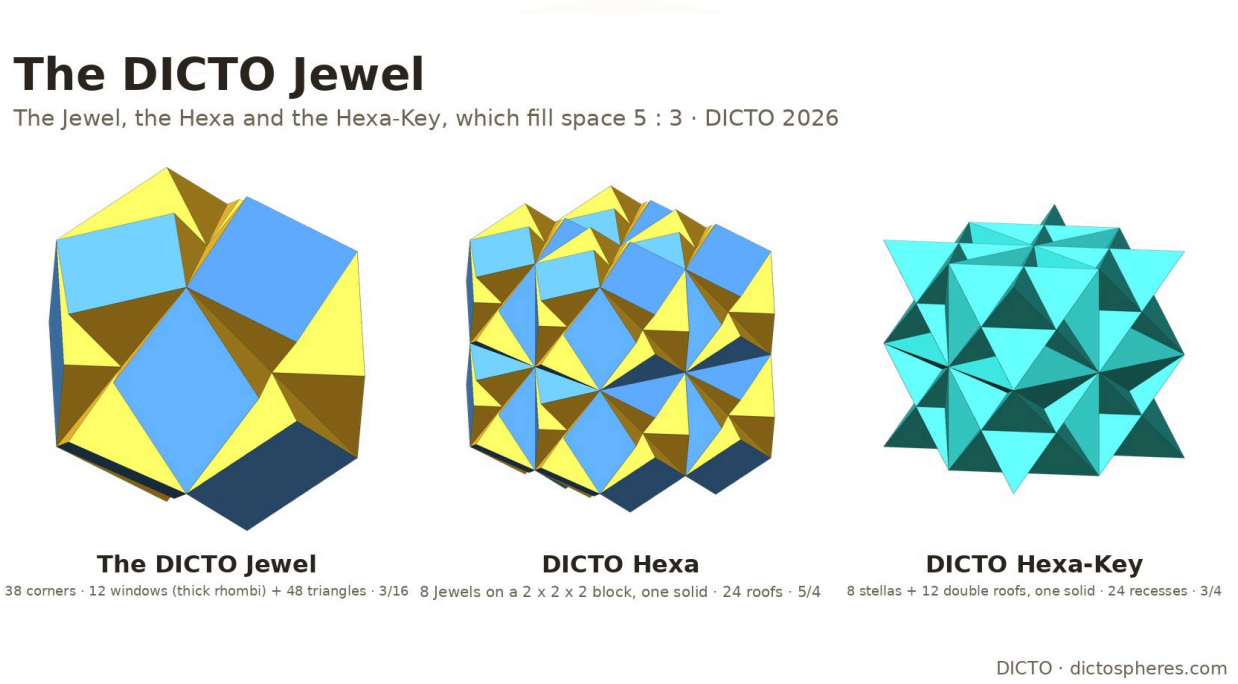
The DICTO Jewel

Euclid's dodecahedron carved by its neighbours' stella octangulas: an exact solid with twelve Penrose windows, its lattice, the DICTO Jewel Tetrahedron and Octahedron and their octet pattern, the rhombohedron, and the DICTO Hexa and Hexa-Key, all exact

DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com
Version 1 · 20 October 2026 · DICTO DOI: reserved on publication · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/570b45a/studies/2026-10-20-dicto-jewel (research branch, commit 570b45a; also attached to this record) · Interactive: dictospheres.com → Kaleidohedra → DICTO → Stella–Jewel Lattice
The Jewel, its stella-octangula counterpart, its name, its clusters (and the Jewel Tetrahedron and Jewel Octahedron as single solids), the Hexa, the Hexa-Key and their names are DICTO's (findings #10, #10b, #15 and #18 on dictospheres.com); computation, exact proofs and drafting with Claude (Anthropic).



Abstract. Put Euclid's dodecahedron on a cube, as roofs on its six faces, and carve away the stella octangulas of the six neighbouring cubes: what remains is the **DICTO Jewel**, with 38 corners and 60 faces, twelve of them thick Penrose rhombi (the windows) and 48 triangles, and volume exactly 3/16 of the Hexa's cube defined below (or 12 on cells of edge 2). With stella octangulas on the other cells, Jewels fill space as a rock-salt pattern, 3 : 1 by volume [2]; we prove this exactly: each stella's cube is the stella and six carved roofs of exactly 1/96 each. DICTO's clusters become single solids: the **DICTO Jewel Tetrahedron** (4 Jewels, 228 faces, 3/4), whose copies share out all Jewels, and the **DICTO Jewel Octahedron** (6 Jewels and the stella that exactly fills their hole, 19/16), which tiles space with loose stellas on exactly 12 lattices of index 12. Together they sit in the pattern of the octet truss, eight Tetrahedra round each Octahedron, but as overlapping solids, not as a tiling: without overlap at most one stella in six can be inside an Octahedron, and the 12 lattices reach that bound. Eight Jewels on the corners of the face-centred cubic cell meet on 18 windows round one stella. The **DICTO Hexa** and **DICTO Hexa-Key** [12] (5/4 and 3/4, filling space 5 : 3, two windows between

> neighbouring Hexas) are proved exactly. As a solid the Jewel is pyritohedral, so it twins like the Ogdoecontahedron [5]; its packing cannot, because a (111) mirror does not keep the cube lattice. All claims are exact in $\mathbb{Q}(\sqrt{5})$ except where marked float.

1. The Jewel (exact)

Frame. As in the EKP cell [1] and [2]: cells are cubes centred on a cubic lattice; a cell is *even* or *odd* by the parity of its index sum, and the even cells form a face-centred cubic lattice. All values below are at the scale where the Hexa's cube (§4) has edge 1, so a cell has edge 1/2 and volume 1/8; the scripts use cells of edge 2, where every volume is 64 times larger.

Construction. Euclid builds the regular dodecahedron on a cube by setting a roof on each face [3]: the hull of the face and a ridge of length $1/\varphi$ times the cell edge, at height $(\varphi - 1)/2$ times the cell edge; each roof is $(1 + \sqrt{5})/192$ and the dodecahedron $(5 + \sqrt{5})/32$. Each roof reaches into the next cell, where the stella octangula (Kepler's two tetrahedra on the cube's corners [4], volume 1/16) takes away $(\sqrt{5} - 1)/192$ of it, leaving a **carved roof** of exactly 1/96. The Jewel is its cell and its six carved roofs: $1/8 + 6/96 =$ **3/16**.

Its surface. 38 corners: the cube's 8, the 12 ridge points, the 6 face centres, and 12 points that complete the windows. Its 60 faces are 12 **windows**, one across each cube edge, lying in the dodecahedron's face planes, and 48 triangles in the stellas' planes; $38 - 96 + 60 = 2$. Each window is a thick Penrose rhombus: side $(\sqrt{5} - 1)/4 = 1/(2\varphi)$, long diagonal 1/2 (the cell's edge), angle 108° at the ridge corners ($\cos 108° = (1 - \sqrt{5})/4$, exact) and 72° at the cube corners [6]. The Jewel's corners are centrosymmetric, have the three-fold turns and the three coordinate mirrors, and no mirror across a (111) plane: it is pyritohedral.

2. The Stella–Jewel Lattice (exact)

Jewels on the even cells and stella octangulas on the odd cells fill space, $3/16 : 1/16 = 3 : 1$ [2]. The first check (finding #10b) sampled points; here it is exact. Take an odd cell. The six Jewels on its face-neighbours each reach into it with one carved roof (their ridges are Jewel corners). These six roofs lie in the cell, any two meet only in a face (the intersection of the uncarved roofs has volume exactly 0), and each carved part is 1/96. With the stella they give $1/16 + 6/96 = 1/8$, the cell: so an odd cell is exactly its stella and six carved roofs, an even cell is the core of its Jewel, and the Jewels and stellas tile space.

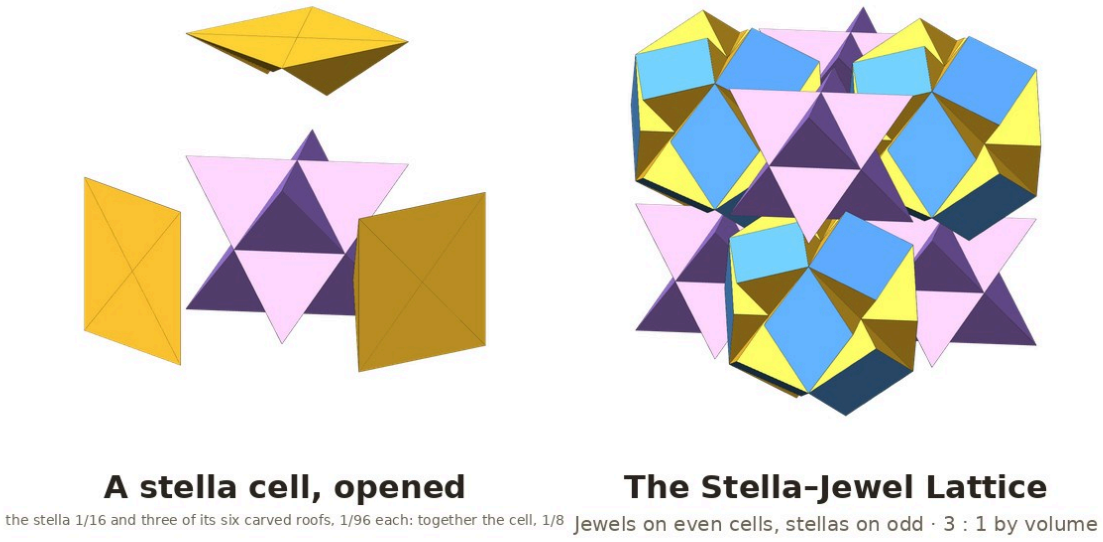


Figure 1. Left: an odd cell opened, its stella octangula (violet) and three of the six carved roofs that the neighbouring Jewels push into it. Right: a 2 × 2 × 2 block of the Stella–Jewel Lattice, drawn slightly apart.

3. Clusters and the rhombohedron (exact)

Neighbours. Two Jewels on neighbouring even cells (one step along a face diagonal) share exactly one window; Jewels two cells apart along an axis share none.

The DICTO Jewel Tetrahedron (DICTO's tetrahedral cluster, finding #15, as one solid): four Jewels on mutually neighbouring even cells. Each pair shares one window, so its surface has 4 × 60 − 2 × 6 = **228 faces** (192 triangles, 36 windows), and all four meet at one point, their cells' common corner; volume 3/4. Every set of four mutually neighbouring Jewels is of this kind, one per cell corner. Anchored on the cells whose three indices are all even, Jewel Tetrahedra use every Jewel exactly once, with the stellas loose between them.

The DICTO Jewel Octahedron (DICTO's octahedral cluster, finding #15, as one solid): the six Jewels round an odd cell, and that cell's stella. Each Jewel shares a window with its four neighbours in the ring (12 windows); inside, the 48 walls of the Jewels' carved roofs enclose a stella-shaped hole, sealed but for its eight tips, and since the odd cell is exactly its stella and the six carved roofs (§2), the stella fills the hole exactly. The solid has an outer surface of **288 faces** (240 triangles, 48 windows) and volume 6 × 3/16 + 1/16 = **19/16**. Jewel Octahedra tile space together with loose stellas on exactly **12 lattices of index 12** (exact integer search over all lattices of that index), for example the one with Hermite basis (1,0,3), (0,1,5), (0,0,12) in cell units; each period holds one Octahedron and five loose stellas.

The octet pattern (DICTO's octet network [10]). The Jewels' cells form a face-centred cubic lattice; its tetrahedral holes are the cell corners and its octahedral holes the odd cells, so Jewel Tetrahedra and Jewel Octahedra sit exactly where the octet truss has its tetrahedra and octahedra. Each Jewel lies in 8 Tetrahedra and 6 Octahedra, and each of an Octahedron's 8 faces (three mutually neighbouring Jewels) is completed to a Tetrahedron by exactly one Jewel outside. It is not an octet tiling: neighbouring Tetrahedra and Octahedra share Jewels,

and what fills space is still the Jewels and the stellas. Without overlap it cannot be one either: each Octahedron takes six Jewels and one stella, and there are as many Jewels as stellas, so at most one stella in six can be inside an Octahedron; the 12 lattices above reach that bound, leaving five stellas in six loose.

The rhombohedron [2]: eight Jewels on the corners of the face-centred cubic cell. Its faces have 60° angles, so the Jewels meet not only along its 12 edges but also along the 6 short face diagonals: **18 windows**. All six face-neighbours of the odd cell at its centre are among its corners, so that cell's stella is sealed inside: 8 Jewels and a stella, 25/16 = 8⅓ Jewels.

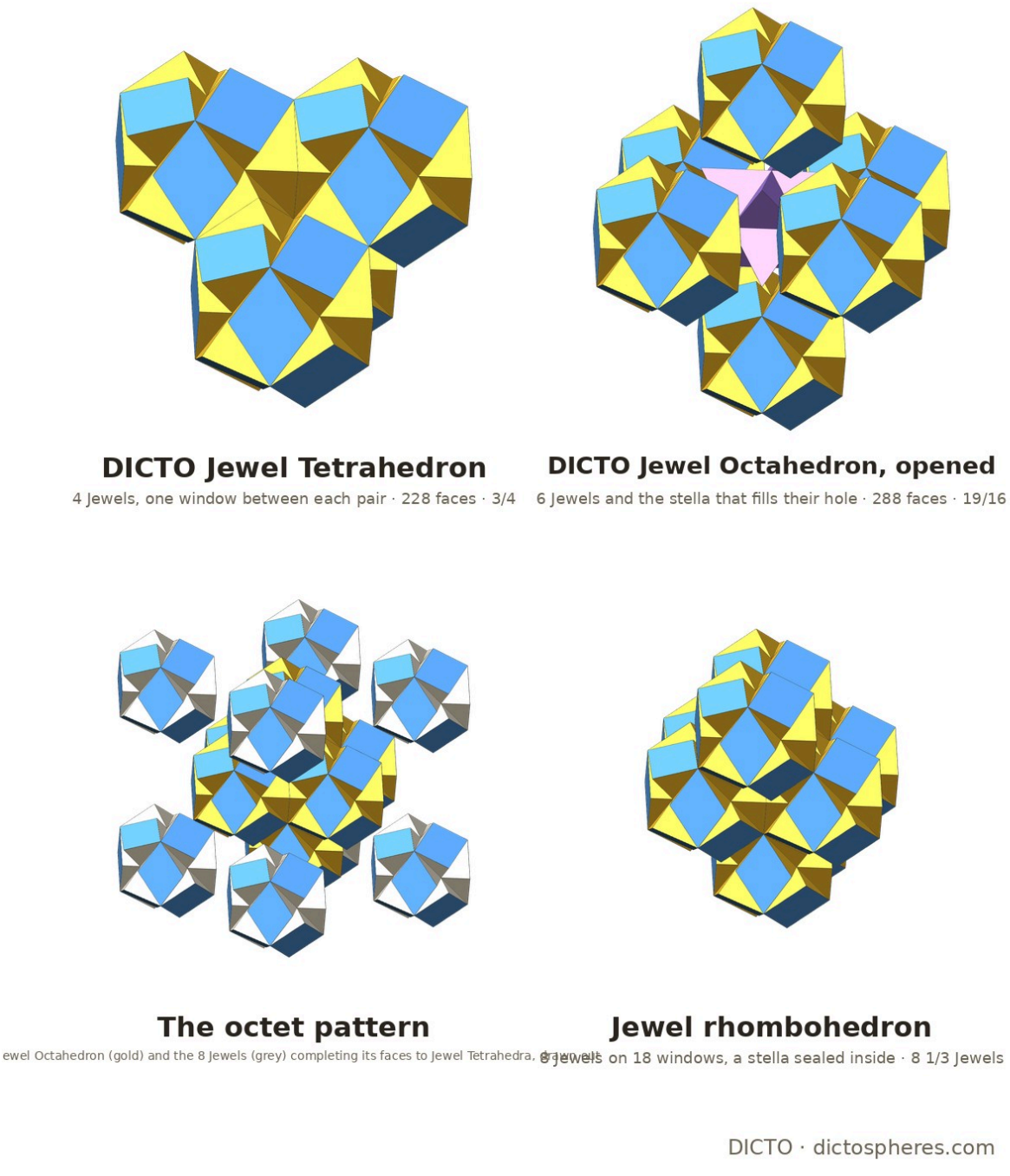


Figure 2. The DICTO Jewel Tetrahedron; the DICTO Jewel Octahedron, its Jewels drawn out to show the stella (violet) that fills its hole; the octet pattern, a Jewel Octahedron (gold) and the 8 Jewels (grey) that complete its faces to Jewel Tetrahedra, drawn out; and the Jewel rhombohedron (windows blue).

4. The DICTO Hexa and the DICTO Hexa-Key (exact)

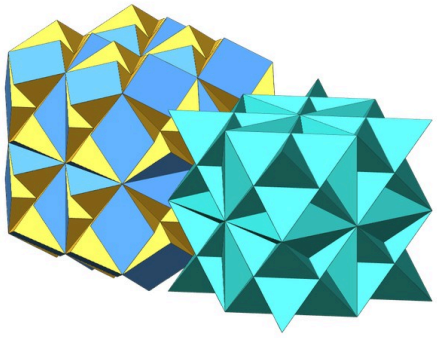
DICTO's Hexa and Hexa-Key and their 5 : 3 filling are published in [12]; here they are proved exactly.

The Hexa. Put a whole Jewel on each of the eight cells of a 2 × 2 × 2 block, on the even and the odd cells alike. Inside the block they overlap; outside it, each of the 24 outer cell faces carries its Jewel's roof. As one solid, the **DICTO Hexa** is the block (edge 1) with 24 carved roofs on its faces: 1 + 24/96 = **5/4**, six and two-thirds Jewels.

The Hexa-Key. Put Hexas on alternate blocks, rock-salt fashion (the blocks of even index sum). Each other block is then entered by 24 roofs, four through each face, from its six neighbouring Hexas; what is left is one solid, the **DICTO Hexa-Key**: 1 − 24/96 = **3/4**, the volume of four Jewels. It is eight stella octangulas, one per cell, joined by twelve double roofs across the inner cell faces: 8/16 + 12 × 2/96 = 3/4. Its 24 recesses are exactly the Hexas' 24 roofs, so Hexas and Hexa-Keys fill space, **5 : 3**.

The same pattern one level up [12]. A cell with six carved roofs added is a Jewel (1/8 + 6/96), with six taken away a stella (1/8 − 6/96); a block with 24 carved roofs added is a Hexa, with 24 taken away a Hexa-Key; and both levels sit in the rock-salt pattern [7].

Windows. The Hexas sit on a face-centred cubic lattice of blocks. Each Hexa meets each of its 12 neighbours on exactly **two whole windows** (24 in all, exact), and no two Hexas' roofs overlap. The Hexa is a union of overlapping Jewels, which is why it is defined as one solid.



Hexa and Hexa-Key
the Hexa's roofs fit the Hexa-Key's recesses · 5 : 3

DICTO · dictospheres.com

Figure 3. A Hexa and the Hexa-Key next to it in their 5 : 3 filling, drawn out a little: the Hexa's roofs go into the Hexa-Key's recesses.

5. Twins

By the twin rule [5], a centrosymmetric solid with a three-fold axis has as its mirror image in a plane at right angles to that axis the same solid turned 60°. The Jewel is such a solid, so as a

solid it twins like the DICTO Ogdoecontahedron. Its packing does not: in [8] a twin through a (111) layer of the Stella–Jewel Lattice left about 4% of sampled points uncovered and 4% covered twice (float). The reason is exact and simple: the mirror in a (111) plane takes the cubic lattice of cells to a different lattice (the step to the next cell, (1, 0, 0) in cell units, goes to (1/3, −2/3, −2/3)), so the mirrored cells cannot meet the cells on the other side face to face. The Ogdoecontahedron's packing twins because its cells, octahedra, are mirrored into the next octahedra of the gyrated octet; the Jewel's cells are cubes, and a mirrored cube does not continue the cubic lattice.

6. Prior art and what is claimed

- **Classical parts.** The dodecahedron as a cube with six roofs is Euclid's (Elements XIII.17) [3]; the stella octangula is Kepler's [4]; the thick rhombus is Penrose's [6]; the rock-salt arrangement is that of sodium chloride [7]; the EKP cell holds them together [1].
- **Lalvani.** US 4,723,382 [9] builds periodic and non-periodic structures from polygonal members whose edges all lie on the fifteen two-fold axes of the icosahedral group. All 48 window edges of the Jewel lie on those axes, but none of its other 48 edges do (lalvani_edges.py, float), so the Jewel, and with it the stella and the lattices here, fall outside that edge condition (our reading, not legal advice).
- **Already published.** The Jewel's definition, its volume and the Stella–Jewel Lattice are in [2], with the cluster networks and the rhombohedral cluster; the Jewel, the lattice, the tetrahedral and octahedral clusters and the octet network were first released with the Kaleidohedra app (findings #10, #10b, #15 [10]); the Hexa, the Hexa-Key, their 5 : 3 filling and the two windows between Hexas are in [12].
- **Claimed here:** the exact proofs of the Jewel's surface, volume and symmetry and of the Stella–Jewel tiling; the DICTO Jewel Tetrahedron and Jewel Octahedron as exact single solids (the stella filling the Octahedron's hole exactly), with the 12 lattices of Octahedra; the octet pattern and its bound of one stella in six; the 18 windows of the rhombohedron; exact proofs of the Hexa, the Hexa-Key, their 5 : 3 filling and the two-window contacts; and the exact reason the packing does not twin.

In the sources checked (the full text of the 80 library sources listed in the appendix, and web searches on 20 October 2026) we did not find the Jewel, its clusters, the Hexa or the Hexa-Key outside DICTO's own records [2, 10, 12].

7. Further work

- A third level: Hexas on a 2 × 2 × 2 block of blocks.
- A physical puzzle: Hexas and Hexa-Keys interlock and can be cut in wood or printed.

Data and reproducibility

Exact (ℚ(√5)): jewel_exact.py (roof, Jewel, Stella–Jewel tiling, Hexa and Hexa-Key, windows between Hexas, the (111) mirror; log jewel-exact.log, 18 PASS) and cluster_exact.py (Jewel Tetrahedron and Octahedron, the 12 lattices, the octet pattern and its bound, the rhombohedron; log cluster-exact.log, 18 PASS), with the exact tools of [11]. Float: lalvani_edges.py. Figures: jewel_figs.py (float copies of the exact pieces), shots.mjs, jewel_compose.py.

References

1. DICTO, The Euclid–Kepler–Pacioli Cell, DICTO DOI: 10.5281/zenodo.23284106.
2. DICTO, Kagome Hulls, DICTO DOI: 10.5281/zenodo.23376317.
3. Euclid, The Thirteen Books of the Elements, tr. T. L. Heath, vol. III (Books X–XIII), Cambridge (1908), Book XIII, Proposition 17.
4. J. Kepler, Harmonices mundi libri V (1619), Book II.
5. DICTO, The DICTO OG Family on Every Stacking, DICTO DOI: 10.5281/zenodo.23490114.
6. R. Penrose, Pentaplexity: a class of non-periodic tilings of the plane, Mathematical Intelligencer 2, 32–37 (1979).
7. W. L. Bragg, The structure of some crystals as indicated by their diffraction of X-rays, Proc. R. Soc. Lond. A 89, 248–277 (1913).
8. DICTO, The DICTO Ogdoecontahedron on the Gyrated Octet, DICTO DOI: 10.5281/zenodo.23484012.
9. H. Lalvani, Building structures based on polygonal members and icosahedral symmetry, US Patent 4,723,382 (1988).
10. DICTO, Kaleidohedra by DICTO, DICTO DOI: 10.5281/zenodo.23173809; releases 10.5281/zenodo.23223555 (#10b) and 10.5281/zenodo.23257393 (#15, #16).
11. DICTO, The DICTO Ogdoecontahedron on the Octet Truss, DICTO DOI: 10.5281/zenodo.23457016 (the exact tools q5.py and exactpoly.py).
12. DICTO, The DICTO Hexa, DICTO DOI: 10.5281/zenodo.23376320.

Appendix. The reference library searched

The full text of the 80 sources below in DICTO's reference library was searched on 20 October 2026 for: Penrose, stella octangula, rock salt, Lalvani, Euclid, roof and rhombic. None contains the results of this paper. The relevant hits are Euclid and Kepler [3, 4], Pacioli, Messer's stellations of the rhombic triacontahedron (the stella octangula), Lalvani's NASA report on octahedral–tetrahedral trusses, and the papers on Penrose tilings (Kramer, Baake and Grimm, Dietl and Eschenburg). Thirteen further items are page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, among them US 4,723,382, read separately in full on 8 October 2026, and Fedorov 1885).

Web searches (20 October 2026): a dodecahedron carved by stella octangulas; space-filling pairs of a dodecahedron-based solid and the stella octangula; cubes of overlapping dodecahedra.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra

S. Kabai	2007	54-Faced Space-Filling Polyhedron (Wolfram Demonstration; description and screenshot)
S. Kabai	2007	Golden Octet Truss (Wolfram Demonstration; description and screenshot)
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2008	A Rhombic Dodecahedron Made of Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
S. Kabai	2008	Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (Wolfram Demonstration; description and screenshot)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
S. Kabai	2015	Cluster of 14 Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb

A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian et al.	2020	Bi-axial woven tiles: interlocking space-filling shapes based on symmetries of bi-axial weaving patterns
V. R. Krishnamurthy, E. Akleman, S. G. Subramanian, M. Ebert, J. Cui, C. Fu, C. Starrett	2022	Geometrically interlocking space-filling tiling based on fabric weaves
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
R. Akpanya et al.	2024	Topologically interlocking blocks inside the tetroctahedrille
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Crystal twinning (incl. the spinel law)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Kyūshindō
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Tetrahedral-octahedral honeycomb (incl. the gyrated form)